Nikhil Mahadevan

Nikhil.Mahadevan@uga.edu • linkedin.com/in/nmahadevan12 • github.com/nmahadevan12 • nmahadevan12.github.io

Education

University of Georgia | Athens, GA **Expected May 2028**
B.S. in Electrical and Electronics Engineering, B.S. in Computer Systems Engineering **GPA: 3.93**
Honors: Eagle Scout**,** UGA Dean's List

Experience

Flight Software Engineer | University of Georgia Small Satellite Research Laboratory **Sep 2026 - Present**

- Developing CubeSat flight software in C++ and NASA F Prime, with Zephyr drivers in C on STM32, for a ~15-person software team
- Reimplemented the legacy 7-state flight state machine with voltage debounce, request prioritization, and safe-shutdown sequencing
- Refactored the legacy file-recycler into thread-safe C++ with path abstractions and safer symlink handling
- Writing unit tests for the state machine and file recycler to catch bugs before system integration
- Reviewing teammates' flight software changes and building UART/RS-485 Zephyr drivers, validated in Renode

Embedded Systems Research Intern | University of Georgia **Aug 2026 - Present**

- Writing C++ drivers (I2C, SPI) to read IMU, temperature, and humidity sensors on an ATmega328P canine health monitor
- Streaming structured multi-sensor packets to a host application in real time over HC-05 Bluetooth via hardware UART

Electronics Lab Assistant | University of Georgia **Aug 2026 - Present**

- Instructing 100+ students across two labs on analog filters, oscillators, transformers and BJT amplifiers with SPICE verification
- Guiding students with TTL circuit builds and benchtop debugging using oscilloscopes, logic analyzers and multimeters

Engineering Peer Tutor | University of Georgia **Sep 2025 - May 2026**

- Tutored 35+ students across courses mainly focused on Logic Design (CSEE 2220) and Circuit Analysis (ECSE 2170)

Projects

Edge AI Wake-Word Engine **Aug 2026**

- Engineered register-level C firmware on an STM32L4 (Cortex-M4) for real-time PDM audio acquisition
- Implemented a DC-blocking filter and energy detector in C to isolate speech buffers and reject background noise
- Developed a 921.6 kbps UART streaming driver with SysTick timestamping to transfer audio buffers to a host PC
- Built a Python tool to read the UART stream and plot the waveform, **verifying** the firmware's audio and noise detection before running the model on the chip
- Trained a "Hey STM" wake-word model (MFCC + logistic regression, 92% accuracy on 200 held-out samples) and ran it on the Cortex-M4 in ~4.5 KB of RAM at ~80 µs per inference

2-Axis Light Tracker **Jul 2026**

- Built a battery-powered dual-axis light tracker in bare-metal C on an ATmega328P, with motor control through MOSFET H-bridges
- Combined an LDR array with Hall-effect and potentiometer feedback to keep tilt within safe limits and stop motor over-rotation
- Wrote a fixed-point Kalman filter that cut ADC noise from 3 LSBs to < 1 LSB without an FPU
- Streamed raw and filtered ADC data, plus I/O data, to a PC over a register-level 115.2 kbps UART driver
- Designed a 4-layer KiCad PCB with separate power, ground, and signal layers, and widened the power traces to reduce voltage drop

Multifunction Instrumentation System **Apr 2026**

- Built a Raspberry Pi digital multimeter and function generator in Python with a ±12 V split-rail supply and active virtual ground
- Designed a SAR ADC from a digital potentiometer and comparator for a 0-5 V voltmeter, 500-10 kΩ ohmmeter, and ±5 V reference
- Implemented function generators (0-10 Vpk, 0-10 kHz) and a 0-10 kHz frequency counter with op-amp conditioning
- Improved DC resolution ~6x and voltmeter precision ~2x by adding digipot calibration LUTs
- Designed a 4-layer mixed-signal Pi HAT PCB in KiCad with dedicated power and ground planes

Technical Skills

Languages: C, C++, Python, ARM Assembly, MATLAB, Java
Embedded Systems: ARM Cortex-M4 (STM32), ARM Cortex-A53 (BCM2837, Raspberry Pi), ATmega328P, ESP32
Firmware & Protocols: Zephyr RTOS, NASA F Prime, UART, SPI, I2C, PWM
Hardware & Tools: KiCad, Altium Designer, SPICE, Fusion 360, Oscilloscopes, Logic Analyzers, Multimeters, Soldering, Renode, Git